

AI Usage Note & Prompt Documentation

Project

SCD2 Copilot – AI-Assisted Slowly Changing Dimension Type 2 Builder

AI Tools Used

- Google Antigravity (primary development assistant)
 - Claude (reasoning and code review)
 - Gemini API (runtime explanation generation)
 - Groq API (fallback explanation provider)
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What AI Helped With

AI was used to accelerate development in the following areas:

- Architecture planning and project structuring
- Streamlit dashboard design and UI improvements
- SCD2 transformation scaffolding
- Validation rule generation
- Unit and integration test generation
- Adversarial dataset creation
- Documentation drafting
- Performance benchmark planning
- Enterprise readiness review

AI significantly reduced development time while enabling rapid iteration and testing.

What AI Got Wrong

Several AI-generated outputs required manual correction:

1. Generated a DuckDB validation query that failed due to row identifier issues.
2. Introduced CSS conflicts that caused Streamlit expanders to display incorrectly.
3. Missed same-day SCD2 update edge cases during early implementation.
4. Incorrectly assessed some local environment dependencies.
5. Required multiple iterations to achieve a professional enterprise dashboard design.

All issues were manually reviewed, corrected, and verified.

Best Prompts Used

Architecture Review

"Review the entire repository, verify all technology choices against official documentation, and ensure deterministic SCD2 processing remains completely independent from LLM-generated content."

SCD2 Logic

"Implement a Slowly Changing Dimension Type 2 transformation that preserves historical records, closes previous active versions correctly, and inserts new active records."

Validation Framework

"Generate validation rules for duplicate active records, overlapping date ranges, null business keys, schema completeness, and date consistency."

Testing

"Create adversarial and property-based tests covering duplicates, schema drift, overlapping validity periods, large change volumes, and edge cases."

Enterprise UI Review

"Redesign the Streamlit application as an internal enterprise operations dashboard without changing any business logic."

Human Validation

All AI-generated code, tests, prompts, documentation, and UI changes were manually reviewed, tested, and verified.

Key verification activities included:

- Manual dataset validation
 - Edge-case testing
 - Adversarial scenario testing
 - UI review
 - Validation rule verification
 - LLM output verification
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Conclusion

AI acted as a development accelerator rather than an autonomous developer. Deterministic SCD2 logic, validation rules, testing strategy, and final acceptance decisions were reviewed and validated by the development team. AI was primarily used to improve productivity, documentation quality, testing coverage, and user experience.